# Intake of dioxins and related compounds from food in the U.S. population.

The first U.S. nationwide food sampling with measurement of dioxins, dibenzofurans, and coplanar, mono-ortho and di-ortho polychlorinated biphenyls (PCBs) is reported in this study. Twelve separate analyses were conducted on 110 food samples divided into pooled lots by category. The samples were purchased in 1995 in supermarkets in Atlanta, GA, Binghamton, NY, Chicago, IL, Louisville, KY, and San Diego, CA. Human milk also was collected to estimate nursing infants' consumption. The food category with highest World Health Organization (WHO) dioxin toxic equivalent (TEQ) concentration was farm-grown freshwater fish fillet with 1.7 pg/g, or parts per trillion (ppt), wet, or whole, weight. The category with the lowest TEQ level was a simulated vegandiet, with 0.09 ppt. TEQ concentrations in ocean fish, beef, chicken, pork, sandwich meat, eggs, cheese, and ice cream, as well as human milk, were in the range O.33 to 0.51 ppt, wet weight. In whole dairy milk TEQ was 0.16 ppt, and in butter 1.1 ppt. Mean daily intake of TEQ for U.S. breast-fed infants during the first year of life was estimated at 42 pg/kg body weight. For children aged 1-11 yr the estimated daily TEQ intake was 6.2 pg/kg body weight. For males and females aged 12-19 yr, the estimated TEQ intake was 3.5 and 2.7 pg/kg body weight, respectively. For adult men and women aged 20-79 yr, estimated mean daily TEQ intakes were 2.4 and 2.2 pg/kg body weight, respectively. Estimated mean daily intake of TEQ declined with age to a low of 1.9 pg/kg body weight at age 80 yr and older. For all ages except 80 yr and over, estimates were higher for males than females. For adults, dioxins, dibenzofurans, and PCBs contributed 42%, 30%, and 28% of dietary TEQ intake, respectively. DDE was also analyzed in the pooled food samples.